IV Infusions Practice 3

Module 3 - Intravenous Infusions · Daily practice sheet · No date

Name _____ Date _____

a A female patient (35 years old, 65 kg) with normal renal function is given a drug by IV infusion with no loading dose. The elimination half-life is 7 hours. How long after the start of the infusion would it take to reach 95% of the steady-state concentration?

b A female patient (35 years old, 65 kg) is to be given a drug by IV infusion. The elimination half-life is 7 hours and the apparent volume of distribution is 23.1% of body weight. The desired steady-state plasma level is 10 mcg/mL. What is an appropriate infusion rate?

c A female patient (35 years old, 65 kg) with normal renal function is to be given a drug by IV infusion. According to the literature, the elimination half-life of this drug is 7 hours and the apparent V_D is 23.1% of body weight. The pharmacokinetics of this drug assumes a first-order process. The desired steady-state plasma level for this antibiotic is 10 mcg/mL. What is an appropriate loading dose for this antibiotic?

e A female patient (35 years old, 65 kg) with normal renal function is to be given a drug by IV infusion. According to the literature, the elimination half-life of this drug is 7 hours and the apparent V_D is 23.1% of body weight. The pharmacokinetics of this drug assumes a first-order process. The desired steady-state plasma level for this antibiotic is 10 mcg/mL. If the drug were infused for 4 hours at the rate you determined above, what would be the concentration of drug in the plasma at the end of the infusion?

f A female patient (35 years old, 65 kg) with normal renal function is to be given a drug by IV infusion. According to the literature, the elimination half-life of this drug is 7 hours and the apparent V_D is 23.1% of body weight. The pharmacokinetics of this drug assumes a first-order process. The desired steady-state plasma level for this antibiotic is 10 mcg/mL. If the drug were infused for 4 hours at the rate you determined above, what would be the concentration of drug in the plasma 6 hours after the cessation of the infusion?

g A female patient (35 years old, 65 kg) with normal renal function is to be given a drug by IV infusion. According to the literature, the elimination half-life of this drug is 7 hours and the apparent V_D is 23.1% of body weight. The pharmacokinetics of this drug assumes a first-order process. The desired steady-state plasma level for this antibiotic is 10 mcg/mL. If the drug were infused long enough to reach steady-state at the rate you determined above, what would be the concentration of drug in the plasma 8 hours after the cessation of the infusion?

i A female patient (35 years old, 65 kg) with normal renal function is to be given a drug by IV infusion. According to the literature, the elimination half-life of this drug is 7 hours and the apparent V_D is 23.1% of body weight. The pharmacokinetics of this drug assumes a first-order process. The desired steady-state plasma level for this antibiotic is 10 mcg/mL. What rate of infusion and loading dose would you recommend to achieve a steady-state concentration of 30 mg/L? (Answer here: the rate of infusion.)

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